

The Bridge to Post-Quantum: Benchmarking Hybrid Cryptographic Implementation

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Submitted to: Dr. Sumaiyah Zahid

Abstract

The advent of fault-tolerant quantum computing threatens the integer-factorization hardness assumptions underpinning RSA and elliptic-curve cryptography. This paper presents a two-part empirical study. In Part 1, we construct gate-level Shor's algorithm circuits using Qiskit for $N \in \{15, 21, 35\}$ and extrapolate qubit and gate-count requirements to RSA-2048, projecting approximately 4,099–6,144 logical qubits and a wall-clock attack time of under five hours on superconducting hardware ($\tau = 1 \mu\text{s}$). In Part 2, we benchmark RSA-2048 KEM, ML-KEM-512 (FIPS 203), and a KDF-based Hybrid KEM across 1,000 runs on consumer hardware. ML-KEM-512 achieves key-generation speedups of approximately $5,700\times$ over RSA-2048, and Hybrid KEM encapsulation adds only 0.07 ms of overhead versus standalone ML-KEM. These results jointly quantify the quantum threat and confirm that a hybrid KEM bridge is both secure and operationally practical for near-term deployment.

Keywords — post-quantum cryptography, ML-KEM, RSA-2048, hybrid KEM, Shor's algorithm, Qiskit, FIPS 203, crypto-agility.

I. INTRODUCTION

The security of virtually all public-key cryptographic infrastructure — from TLS certificates and digital signatures to key-encapsulation mechanisms — rests on the computational hardness of integer factorization (RSA) and the discrete logarithm problem (ECC). Peter Shor's 1994 polynomial-time quantum algorithm dissolves both assumptions, reducing the factoring of an n -bit modulus to $O(n^3)$ quantum gate operations [2].

No fault-tolerant quantum computer exists today that could break RSA-2048 — but the gap is closing faster than most migration timelines assume. Google's Willow chip achieving below-threshold error correction (December 2024), IonQ's 2028 roadmap, and Quantinuum's fault-tolerant trapped-ion demonstrations — indicate that the transition window is measured in years, not decades. The "harvest now, decrypt later" (HNDL) threat model compounds urgency: adversaries may already be archiving TLS-encrypted traffic for future decryption.

In August 2024, NIST released three post-quantum standards: FIPS 203 (ML-KEM, derived from CRYSTALS-Kyber), FIPS 204 (ML-DSA), and FIPS 205 (SLH-DSA) [3]. However, a direct cutover from RSA to PQC is operationally infeasible for most organizations, which face 12–15+ year migration timelines. Hybrid KEMs — which combine a classical and a post-quantum primitive so that security holds if either component remains unbroken — provide the pragmatic bridge.

Despite strong standardization guidance, the published literature lacks (a) open, gate-level Shor threat models with extrapolations to RSA-2048, and (b) reproducible, statistically robust benchmark data on consumer hardware. This paper closes both gaps with a two-part study: a Qiskit Shor circuit analysis

IV. RESULTS

A. Shor Circuit Projections

Table I summarizes simulated and projected qubit and gate requirements. For $N = 15$ ($n = 4$), the circuit requires 12 logical qubits and approximately 128 gates. Fitting a $3n^3$ model across the three data points yields the RSA-2048 projection.

TABLE I: Shor Circuit Projections

N	Bits (n)	Qubits	Gates (approx.)	Model
15	4	12	128	Sim.
21	5	15	250	Sim.
35	6	18	432	Sim.
2^{2048}	2048	4,099–6,144	1.72×10^{10}	Extrap.

The empirical (naïve) projection yields 6,144 logical qubits; the optimized theoretical model yields 4,099, consistent with prior literature estimates of approximately 4,000 logical qubits. Runtime projections for RSA-2048: ≈ 4.77 hours on superconducting hardware ($\tau = 1 \mu\text{s}$ gate time) and ≈ 19.88 days on trapped-ion hardware ($\tau = 100 \mu\text{s}$). Both figures assume logical (error-corrected) gate times; physical qubit overhead from surface codes would increase these estimates substantially.

B. PQC Benchmark Results

Table II presents 1,000-run benchmark results across the three KEM schemes.

TABLE II: 1,000-Run KEM Benchmark (mean $\pm \sigma$, ms)

Operation	RSA-2048	ML-KEM-512	Hybrid KEM
KeyGen (ms)	1150 ± 12.30	0.0151 ± 0.0034	—

(Part 1) and a 1,000-run empirical benchmark of RSA-KEM, ML-KEM-512, and Hybrid KEM (Part 2).

II. BACKGROUND & RELATED WORK

A. Shor's Algorithm and the RSA Threat

Shor's algorithm reduces RSA-n factoring to quantum period-finding via the Quantum Fourier Transform (QFT). The register size requires $t = 2m$ control qubits ($m = \lceil \log_2 N \rceil$) plus m target qubits, yielding $3m$ total logical qubits as a lower bound. Gidney and Ekerå (2021) estimate 20 million noisy physical qubits to break RSA-2048, noting a monotonically decreasing trajectory from 2015 projections, consistent with our empirical model [2].

B. NIST PQC Standardization

Following an eight-year global competition, NIST finalized ML-KEM (FIPS 203) as the primary key-encapsulation standard. ML-KEM-512, -768, and -1024 offer security levels corresponding to AES-128, AES-192, and AES-256 respectively. NIST IR 8547 sets RSA-2048 deprecation for 2030 and full disallowance by 2035 [3].

C. Hybrid KEMs and Crypto-Agility

NIST CSWP 39 (2025) formalizes crypto-agility as the capacity to replace cryptographic primitives without disrupting running systems. RFC 9180 (HPKE) and emerging IETF drafts formalize hybrid KEM composition. Certificate Authorities began issuing hybrid X.509 certificates in 2025, while Google Chrome deployed X25519+Kyber for TLS in 2024 — validating the hybrid bridge approach in production.

III. METHODOLOGY

A. Shor Circuit Construction

We implement Shor's algorithm in Qiskit 1.x for composite integers $N \in \{15, 21, 35\}$ ($n = 4, 5, 6$ bits). The circuit comprises three stages:

(1) QFT/IQFT: The Quantum Fourier Transform uses Hadamard gates combined with controlled-phase rotations ($\theta = 2\pi / 2^k$ for depth k). Bit-reversal swaps enforce standard ordering; IQFT uses conjugate-negative rotations.

(2) Modular Unitaries: The modular exponential oracle U_a encodes controlled permutation unitaries $U_a^{\{2^i\}}$ over the target register. Register sizing follows $m = \lceil \log_2 N \rceil$ for the work register and $t = 2m$ for the counting register.

(3) Post-Processing: After measurement of the counting register, the observed bit-string is used to estimate the period r via continued-fraction expansion. Prime factors are then recovered via $g_1 = \gcd(a^{\{r/2\}} - 1, N)$ and $g_2 = \gcd(a^{\{r/2\}} + 1, N)$.

Encap (ms)	0.1705 ± 0.0133	0.0716 ± 0.0076	0.2443 ± 0.0189
Decap (ms)	1.9961 ± 0.0811	0.0673 ± 0.0069	~ 2.06

ML-KEM-512 is approximately $5,700\times$ faster than RSA-2048 at key generation, $2.4\times$ faster at encapsulation, and $30\times$ faster at decapsulation. The hybrid scheme's encapsulation latency of 0.2443 ms represents the additive cost of running both primitives in parallel — approximately 0.07 ms over standalone ML-KEM — confirming negligible overhead.

C. Transmission Size Analysis

Table III compares transmission sizes relevant to protocol integration.

TABLE III: Transmission Size Comparison (bytes)

Scheme	Pub. Key (B)	Ciphertext (B)	Total (B)
RSA-2048	294	256	550
ML-KEM-512	800	768	1,568
Hybrid KEM	1,094	1,024	2,118

ML-KEM-512's larger key and ciphertext sizes (800 B and 768 B respectively) reflect its lattice-based structure. The Hybrid KEM's total transmission footprint of approximately 2.1 KB represents a manageable increase for TLS handshakes and certificate exchanges — well within the constraints of modern network MTUs.

V. DISCUSSION

A. Threat Quantification and Implications

Our Shor projections confirm that breaking RSA-2048 requires approximately 4,099 logical qubits — a target reachable within current decade hardware roadmaps. The 4.77-hour superconducting runtime assumes error-free logical qubits; accounting for surface-code overhead (approximately 1,000 physical qubits per logical qubit at current error rates) raises the physical qubit requirement toward the millions, consistent with Gidney and Ekerå [2]. However, rapid progress in error correction — exemplified by Google Willow's below-threshold demonstration — compresses this timeline in ways that 2021 estimates did not anticipate.

The HNDL threat, however, makes the question of when FTQCs arrive less critical than when migration is complete. Organizations archiving TLS traffic today need RSA-2048 deprecated before decryption becomes feasible — not after.

B. Hybrid KEM as a Practical Bridge

Our benchmarks show that the hybrid KEM construction imposes latency overhead of approximately 0.07 ms at encapsulation relative to standalone ML-KEM-512. This is operationally

Gate counts and qubit requirements are logged per simulation and fitted to derive extrapolation models for $N = 2^{2048}$ (RSA-2048).

B. PQC Benchmarking

All benchmarks execute on consumer hardware (Intel Core i7, 16 GB RAM) under isolated single-threaded conditions. Three KEMs are evaluated: (i) RSA-2048 KEM via PKCS#1 OAEP wrapping a 32-byte random secret; (ii) ML-KEM-512 from liboqs compiled from source; and (iii) a Hybrid KEM that runs RSA and ML-KEM in parallel, concatenates public keys and ciphertexts, and derives the shared secret as $\text{SHA-256}(\text{ss_RSA} \parallel \text{ss_ML-KEM})$.

Each of the three operations — KeyGen, Encapsulation, and Decapsulation — is measured over 1,000 independent runs using high-resolution wall-clock timing at microsecond precision after a warmup phase. Results are reported as mean $\pm$ one standard deviation.

indistinguishable from RSA-only latency in any network context and is negligible relative to typical TLS handshake round-trip times (≥ 10 ms). The dual-security property — that the hybrid scheme is broken only if both RSA and ML-KEM are simultaneously broken — provides a stronger security guarantee than either alone during the transition window.

The 2.1 KB total transmission footprint of the hybrid scheme warrants attention in highly constrained environments (IoT, satellite links), but is unproblematic for enterprise TLS, VPN, and API contexts, which this study targets.

C. Limitations

This study benchmarks on a single hardware configuration; multi-core, ARM, and constrained-device results may differ. The Shor simulation uses ideal (noiseless) Qiskit statevector simulation; real device execution with error mitigation is beyond this study's scope. Physical qubit overhead from error correction is not modeled.

VI. CONCLUSION & FUTURE WORK

This paper provides a joint empirical answer to two open questions in the PQC migration literature: how quantifiable is the Shor threat to RSA-2048, and how practical is a hybrid KEM bridge?

On the threat side, our gate-level Shor models project RSA-2048 vulnerability at approximately 4,099–6,144 logical qubits and under five hours on superconducting hardware — a near-term risk under current hardware trajectories. On the migration side, ML-KEM-512 outperforms RSA-2048 by 5,700 $\times$ in key generation and 30 $\times$ in decapsulation, and a KDF-based Hybrid KEM adds only 0.07 ms of encapsulation overhead, confirming immediate deployability.

Together, these findings support a two-phase transition strategy: (1) deploy Hybrid KEM immediately to achieve HNDL protection and dual-security guarantees while maintaining backward compatibility, and (2) deprecate the RSA component on a planned schedule aligned with NIST IR 8547's 2030 deadline.

Future work will model physical error-correction overhead using surface codes, test PQC performance under constrained network topologies and IoT hardware, evaluate ML-KEM-768 and ML-KEM-1024 for higher-security environments, investigate multithreaded Hybrid KEM encapsulation to reduce sequential overhead, and extend the benchmark methodology to aggregate results across multiple independent sessions — reporting minimum, maximum, and mean-of-means to capture session-to-session variability that single-run evaluations cannot expose.

VII. REFERENCES

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